

Homotopy Pushouts and Mayer-Vietoris

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1 Homotopy Pushout Squares

A homotopy pushout retains a homotopy between the two maps out of the common space. The double mapping cylinder makes this gluing data explicit.

1.1 Double Mapping Cylinders and Their Universal Property

Write $I = [0, 1]$. For a space W , write $\text{pr}_W : W \rightarrow *$ for its unique map to a point. We use Top for the category of topological spaces and continuous maps. As in Lecture 5, homotopy pushout means that the comparison is a homotopy equivalence.

Definition 1.1. Let $f : W \rightarrow X$ and $g : W \rightarrow Y$ be continuous maps. Their *double mapping cylinder* is the quotient

$$\text{Cyl}(f, g) = (X + (W \times I) + Y) / ((w, 0) \sim f(w), (w, 1) \sim g(w)).$$

Let $\iota_X : X \rightarrow \text{Cyl}(f, g)$ and $\iota_Y : Y \rightarrow \text{Cyl}(f, g)$ be the canonical maps. The map $\eta : W \times I \rightarrow \text{Cyl}(f, g)$, given by $\eta(w, t) = [w, t]$, is a homotopy from $\iota_X f$ to $\iota_Y g$.

Proposition 1.2. For continuous maps $f : W \rightarrow X$ and $g : W \rightarrow Y$, and a space Z , continuous maps $\text{Cyl}(f, g) \rightarrow Z$ correspond bijectively to triples (i, j, h) consisting of maps $i : X \rightarrow Z, j : Y \rightarrow Z$, and a homotopy

$$h : W \times I \rightarrow Z, \quad h(w, 0) = if(w), \quad h(w, 1) = jg(w).$$

The correspondence is natural in Z .

Proof. A map on the disjoint union in Definition 1.1 is specified by its three restrictions. It descends through the quotient if and only if the restrictions agree on the identified end-points, which are the displayed conditions on h . The quotient topology gives continuity of the descended map. Uniqueness follows from surjectivity of the quotient map, and postcomposition proves naturality. $\square$

Definition 1.3. Consider a square of continuous maps, together with a specified homotopy $h : if \simeq jg$:

$$\begin{array}{ccc} W & \xrightarrow{f} & X \\ g \downarrow & & \downarrow i \\ Y & \xrightarrow{j} & Z. \end{array}$$

Its *comparison map* is the map $\theta_h : \text{Cyl}(f, g) \rightarrow Z$ from Proposition 1.2, with restrictions i, j, h . The square is a *homotopy pushout* if θ_h is a homotopy equivalence.

The specified homotopy is part of this definition. Even when $if = jg$, a nonconstant chosen homotopy can change the comparison map. For a strictly commutative square, the constant homotopy is the usual choice.

A map out of the ordinary pushout $X \underset{W}{+} Y$ requires equality of the two composites. A map out of $\text{Cyl}(f, g)$ requires a chosen homotopy. Passing to homotopy classes therefore does not turn this construction into an ordinary pushout in the homotopy category: the homotopy between the composites remains part of the mapping data.

Example 1.4. Take $X = Y = *$, where $*$ is a one-point space. For the two maps pr_W , the double mapping cylinder $\text{Cyl}(\text{pr}_W, \text{pr}_W)$ is the *unreduced suspension*

$$SW = (W \times I) / ((w, 0) \sim *_0, (w, 1) \sim *_1).$$

Here $*_0$ and $*_1$ are distinct vertices. In particular, SS^0 is a circle: its two intervals have the same two endpoints. The ordinary pushout of $* \xleftarrow{\text{pr}_{S^0}} S^0 \xrightarrow{\text{pr}_{S^0}} *$ is a point. Thus an ordinary pushout need not be a homotopy pushout. More generally, $SS^m \cong S^{m+1}$ for $m \geq 0$.

1.2 Cofibrations and Geometric Gluing

A Hurewicz cofibration permits a homotopy on a subspace to extend to the whole space. This extension makes ordinary gluing agree with homotopy gluing. The comparison below is the double-mapping-cylinder lemma of May, Chapter 10, §7, p. 80; we give the extension argument explicitly.

Definition 1.5. A continuous map $f : X \rightarrow Y$ is a *Hurewicz cofibration* if it has the homotopy extension property: for every space Z , every map $h : Y \rightarrow Z$, and every homotopy $\alpha : X \times I \rightarrow Z$ starting at hf , there is a homotopy $\beta : Y \times I \rightarrow Z$ starting at h satisfying $\beta(f(x), t) = \alpha(x, t)$ for all $(x, t) \in X \times I$.

For a subspace inclusion, equivalently, the homotopy extension problem is the extension in the diagram

$$\begin{array}{ccc} (Y \times \{0\}) & \xrightarrow{X \times \{0\}} & (X \times I) \longrightarrow Z \\ \downarrow & \nearrow \beta & \uparrow \\ Y \times I & & \end{array}$$

for every continuous map on the upper-left subspace. Inclusions of CW subcomplexes are Hurewicz cofibrations. This standard fact follows by extending over each attached disk; see May, Chapter 6, §1, and Chapter 10, §1.

Proposition 1.6. Let $f : W \hookrightarrow X$ be a Hurewicz cofibration and $g : W \rightarrow Y$ any continuous map. Then the quotient map

$$q : \text{Cyl}(f, g) \rightarrow X \underset{W}{+} Y$$

given on the cylinder by $[w, t] \mapsto [f(w)] = [g(w)]$ is a homotopy equivalence. Consequently, the ordinary pushout square, with its constant homotopy, is a homotopy pushout.

Proof. First form the mapping cylinder $\text{Cyl}(f) = (X + W \times I) / ((w, 0) \sim f(w))$. Let $\iota : X \rightarrow \text{Cyl}(f)$ be the inclusion, $k : W \rightarrow \text{Cyl}(f)$ the map $k(w) = [w, 1]$, and $p : \text{Cyl}(f) \rightarrow X$ the map that is the identity on X and sends $[w, u]$ to $f(w)$.

Extend the homotopy $[w, t]$ from W to a homotopy $F : X \times I \rightarrow \text{Cyl}(f)$ starting at ι , using Definition 1.5. Its terminal map $s(x) = F(x, 1)$ satisfies $sf = k$. The homotopy pF goes from id_X to ps and is stationary on W . A homotopy K from $\text{id}_{\text{Cyl}(f)}$ to sp is given by

$$K(x, t) = F(x, t), \quad K([w, u], t) = [w, u + t(1 - u)].$$

The formulas agree at $u = 0$ because $F(f(w), t) = [w, t]$. They fix $k(W)$ because the second formula has height 1 when $u = 1$. Thus p and s are homotopy inverses under W .

Now glue Y to $\text{Cyl}(f)$ along k and to X along f . The maps p, s descend to maps between $\text{Cyl}(f, g)$ and $X \underset{W}{+} Y$, and the two homotopies descend by taking the identity on Y . The map induced by p is q , so q is a homotopy equivalence. $\square$

Example 1.7. For a map $f : S^{m-1} \rightarrow X$, attaching an m -cell is the pushout

$$\begin{array}{ccc} S^{m-1} & \xrightarrow{f} & X \\ g \downarrow & & \downarrow i \\ D^m & \xrightarrow{j} & X \underset{S^{m-1}}{+} D^m. \end{array}$$

The inclusion $g : S^{m-1} \hookrightarrow D^m$ is a Hurewicz cofibration. Proposition 1.6, with the two legs interchanged, makes this a homotopy pushout. The same conclusion holds for gluing CW complexes along a common subcomplex. Taking both legs to be $S^{m-1} \hookrightarrow D^m$ gives S^m , obtained by gluing two disks along their boundary.

Example 1.8. For $f : W \rightarrow X$, the space $\text{Cyl}(f, \text{pr}_W)$ is the *mapping cone*

$$\text{Cone}(f) = (X + (W \times I)) / ((w, 0) \sim f(w), (w, 1) \sim *).$$

Thus the square $W \rightarrow X, * \rightarrow \text{Cone}(f)$, with its cylinder homotopy, is a homotopy pushout. If W is nonempty and f is a Hurewicz cofibration, Proposition 1.6 gives $\text{Cone}(f) \simeq X/W$, where X/W is defined by $w \sim *$ for all $w \in W$. For the identity map of W , this cone is contractible. When $W = \emptyset$, the mapping cone is $X + *$.

2 Singular Chains and Mayer-Vietoris

The double mapping cylinder has an open cover whose pieces retract to X, Y and whose intersection retracts to W . Small singular chains turn this cover into a short exact sequence and a homotopy pushout of chain complexes.

2.1 Small Chains and the Connecting Homomorphism

For a space Z , write $C_*(Z)$ for its singular chain complex with integer coefficients. If $Z = U \cup V$ is an open cover, let $C_*(U) + C_*(V)$ denote the subcomplex of $C_*(Z)$ generated by singular simplices whose images lie in U or V . Faces of such simplices remain in the same set, so this is a subcomplex.

The following is the small-chain theorem of Hatcher, Proposition 2.21, pp. 119–123, restricted to a cover by two open sets. Its subdivision construction will also be used for excision.

Theorem 2.1. For an open cover $Z = U \cup V$, the inclusion

$$C_*(U) + C_*(V) \hookrightarrow C_*(Z)$$

is a chain homotopy equivalence and hence induces an isomorphism in every homology degree.

The geometric input is that iterated barycentric subdivision makes each simplex in a finite chain lie in a cover member. Pull the cover back to the compact standard simplex and use a Lebesgue number; subdivision decreases diameters by a factor at most $n/(n+1)$ in dimension $n > 0$. The subdivision map is chain homotopic to the identity through a homotopy supported in each original simplex. Thus subdividing a cycle gives a small homologous cycle, and subdividing a bounding chain preserves a small boundary up to a small chain homotopy. These two conclusions give surjectivity and injectivity on homology, respectively.

Let $f : U \cap V \hookrightarrow U$, $g : U \cap V \hookrightarrow V$, $i : U \hookrightarrow Z$, and $j : V \hookrightarrow Z$ be the inclusions. There is a short exact sequence of chain complexes

$$0 \rightarrow C_*(U \cap V) \xrightarrow{(f_*, g_*)} C_*(U) \oplus C_*(V) \xrightarrow{j_* - i_*} C_*(U) + C_*(V) \rightarrow 0. \quad (1)$$

The first map is injective because inclusions preserve the basis of singular simplices. In the middle, the equation $y - x = 0$ implies that $x = y$ is supported on simplices lying in both U and V . This identifies the kernel with the image of the first map. The last map is surjective by definition of small chains.

Theorem 2.2. For an open cover $Z = U \cup V$, the inclusions induce a natural long exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(U \cap V) &\xrightarrow{(f_*, g_*)} H_n(U) \oplus H_n(V) \xrightarrow{j_* - i_*} H_n(Z) \\ &\xrightarrow{\delta_n} H_{n-1}(U \cap V) \rightarrow \cdots \rightarrow H_0(Z) \rightarrow 0. \end{aligned}$$

If a class is represented by a small cycle $z = y - x$, with $x \in C_n(U)$ and $y \in C_n(V)$, then $\delta_n[z] = [\partial x] = [\partial y]$ in $H_{n-1}(U \cap V)$.

Proof. Apply the long exact homology sequence from Lecture 2 to (1), and identify its last complex with $C_*(Z)$ on homology by Theorem 2.1. For a small cycle $z = y - x$, the boundary of its lift (x, y) is $(\partial x, \partial y) = (\partial x, \partial x)$. Exactness of (1) places ∂x in $C_{n-1}(U \cap V)$; the connecting map is its homology class. The construction of the long exact sequence makes this independent of the lift and representative. Maps carrying U, V into the corresponding cover members induce maps of short exact sequences, proving naturality. $\square$

The second map means $(x, y) \mapsto j_*(y) - i_*(x)$. Its composite with (f_*, g_*) is zero because $if = jg$.

2.2 The Cylinder Cover and the Derived Pushout

Let $f : W \rightarrow X$ and $g : W \rightarrow Y$ be continuous maps. Consider the subspaces

$$U = X \times_{W \times \{0\}}^+ (W \times [0, 2/3]), \quad V = (W \times (1/3, 1]) \times_{W \times \{1\}}^+ Y$$

of $\text{Cyl}(f, g)$, using the identifications in Definition 1.1. Their preimages in the defining disjoint union are open and saturated, so U, V are open in $\text{Cyl}(f, g)$. They cover $\text{Cyl}(f, g)$, and $E = U \cap V$ is $W \times (1/3, 2/3)$.

Let $p_U : U \rightarrow X$ and $p_V : V \rightarrow Y$ be the identity on X and Y , respectively, with $p_U([w, t]) = f(w)$ and $p_V([w, t]) = g(w)$ on the cylinder portions. They are deformation retractions, by moving heights linearly to 0 and 1. Projection $p_E : E \rightarrow W$ is a homotopy equivalence. These maps

give the commuting diagram of spans

$$\begin{array}{ccccc} U & \longleftrightarrow & E & \hookrightarrow & V \\ p_U \downarrow & & \downarrow p_E & & \downarrow p_V \\ X & \xleftarrow{f} & W & \xrightarrow{g} & Y \end{array}$$

where the vertical maps on the left and right are $p_U : U \rightarrow X$ and $p_V : V \rightarrow Y$, respectively.

Theorem 2.3. For a homotopy pushout square (f, g, i, j, h) as in Definition 1.3, singular homology has a long exact sequence

$$\begin{aligned} \cdots \rightarrow H_n(W) &\xrightarrow{(f_*, g_*)} H_n(X) \oplus H_n(Y) \xrightarrow{j_* - i_*} H_n(Z) \\ &\xrightarrow{\delta_n^Z} H_{n-1}(W) \rightarrow \cdots \rightarrow H_0(Z) \rightarrow 0. \end{aligned} \quad (2)$$

It is natural for maps preserving the square and its specified homotopy.

Proof. Apply Theorem 2.2 to the cylinder cover. Homotopy invariance from Lecture 6 identifies the homology of U, V, E with that of X, Y, W . The commuting diagram of spans identifies the first map with (f_*, g_*) and the second with $\iota_{Y,*} - \iota_{X,*}$. This gives the sequence for $\text{Cyl}(f, g)$.

The comparison $\theta_h : \text{Cyl}(f, g) \rightarrow Z$ induces an isomorphism on homology. Transport the sequence along it. Since $\theta_h \iota_X = i$ and $\theta_h \iota_Y = j$, the middle map becomes $j_* - i_*$. If $\delta_n^{\text{Cyl}(f, g)} : H_n(\text{Cyl}(f, g)) \rightarrow H_{n-1}(W)$ denotes the cylinder connecting map, then

$$\delta_n^Z = \delta_n^{\text{Cyl}(f, g)} (\theta_{h,*})^{-1}.$$

Naturality of the cylinder cover sequence and of the comparison gives the asserted naturality. $\square$

Theorem 2.3 records the preservation statement at the level of homology groups. At the level of chain complexes it takes the form of a homotopy pushout in the derived category $\mathcal{D}(\mathbb{Z})$, the category of chain complexes with quasi-isomorphisms inverted.

Definition 2.4. For chain maps $f_\# : C_\bullet(W) \rightarrow C_\bullet(X)$ and $g_\# : C_\bullet(W) \rightarrow C_\bullet(Y)$, their *algebraic double mapping cylinder*, or *homotopy pushout*, is the complex $D_\bullet$ with degree n group and differential

$$\begin{aligned} D_n &= C_n(X) \oplus C_n(Y) \oplus C_{n-1}(W), \\ d_D(x, y, w) &= (\partial x + f_\# w, \partial y + g_\# w, -\partial w). \end{aligned}$$

The chain-map identities imply $d_D^2 = 0$. The structure maps are $i(x) = (-x, 0, 0)$ and $j(y) = (0, y, 0)$. The degree-one map $T(w) = (0, 0, w)$ satisfies

$$d_D T + T \partial = j g_\# - i f_\#.$$

Let $\mathbb{Z}[\Delta_{\text{nd}}^1]$ be the interval chain complex with generators e_0, e_1 in degree zero and e_{01} in degree one, and differential $\partial e_{01} = e_1 - e_0$. The cylinder chain map

$$h : C_\bullet(W) \otimes \mathbb{Z}[\Delta_{\text{nd}}^1] \rightarrow D_\bullet$$

is given, for $w \in C_n(W)$, by

$$h(w \otimes e_0) = i f_\#(w), \quad h(w \otimes e_1) = j g_\#(w), \quad h(w \otimes e_{01}) = (-1)^n T(w).$$

Thus h has endpoints $i f_\#$ and $j g_\#$, while T is its degree-one component with the displayed sign. With the mapping-cone convention of Lecture 2, this explicitly defined complex is $\text{Cone}(u)$ for $u = (f_\#, g_\#)$.

Proposition 2.5. For a homotopy pushout square of spaces as in Theorem 2.3, the complex $C_\bullet(Z)$ is isomorphic in $\mathcal{D}(\mathbb{Z})$ to the complex $D_\bullet$ of Definition 2.4. The resulting cone triangle induces (2) on homology.

Proof. For the open cylinder cover, write $u_E : C_\bullet(E) \rightarrow C_\bullet(U) \oplus C_\bullet(V)$ for the pair of inclusions. Its cone maps to small chains by $(x, y, w) \mapsto y - x$. This is a surjective chain map. By (1), its kernel identifies with the cone of the identity of $C_\bullet(E)$, which is contractible. Thus

$$\text{Cone}(u_E) \rightarrow C_\bullet(U) + C_\bullet(V) \rightarrow C_\bullet(\text{Cyl}(f, g))$$

are quasi-isomorphisms, the second by Theorem 2.1.

The retractions p_U, p_V, p_E induce a map of spans of chain complexes. Each is a quasi-isomorphism by homotopy invariance. The induced map of cones $\text{Cone}(u_E) \rightarrow D_\bullet$ is therefore a quasi-isomorphism, by the long exact cone sequences and the five lemma. Together with $C_\bullet(\text{Cyl}(f, g)) \rightarrow C_\bullet(Z)$ induced by θ_h , these give a zigzag of quasi-isomorphisms from $D_\bullet$ to $C_\bullet(Z)$.

The triangle

$$C_\bullet(W) \xrightarrow{(f_\#, g_\#)} C_\bullet(X) \oplus C_\bullet(Y) \rightarrow D_\bullet \rightarrow C_\bullet(W)[1]$$

is the mapping-cone triangle from Lecture 2. Here $C_\bullet(W)[1]_n = C_{n-1}(W)$, with differential $-\partial$. The preceding maps identify its long exact homology sequence with the cylinder-cover sequence and its transport to Z . $\square$

For example, the kernel complex in this proof has differential $(v, w) \mapsto (\partial v + w, -\partial w)$, and the degree-one map $(v, w) \mapsto (0, v)$ contracts it. This supplies the acyclicity used in Proposition 2.5.

3 Generalized Homology and Calculations

The Mayer-Vietoris axiom of Lecture 5 is the abstract form of Theorem 2.3. Suspensions, cell attachments, and mapping cones give its basic computations.

3.1 The Gluing Axiom and Its Meaning

Let $A_n : \text{Top} \rightarrow \text{Ab}$, for $n \in \mathbb{Z}$, be functors equipped with natural Mayer-Vietoris connecting maps for homotopy pushout squares. Assume homotopy invariance, additivity, and exactness of the Mayer-Vietoris sequences. Homotopy invariance sends homotopy equivalences to isomorphisms; additivity identifies $A_n(X + Y)$ with $A_n(X) \oplus A_n(Y)$.

Proposition 3.1. For a homotopy pushout square (f, g, i, j, h) of topological spaces, there is an exact sequence in every integer degree

$$\begin{aligned} \cdots \rightarrow A_n(W) &\xrightarrow{(f_*, g_*)} A_n(X) \oplus A_n(Y) \xrightarrow{j_* - i_*} A_n(Z) \\ &\xrightarrow{\delta_n^Z} A_{n-1}(W) \rightarrow \cdots \end{aligned}$$

If $\delta_n^{\text{Cyl}(f, g)}$ is the connecting map for the span (f, g) , then $\delta_n^Z = \delta_n^{\text{Cyl}(f, g)}(\theta_{h,*})^{-1}$.

Proof. The Mayer-Vietoris sequence of Lecture 5 is put in the displayed convention by multiplying the $A_n(Y)$ summand and the $A_n(\text{Cyl}(f, g))$ term by -1 . Thus the cylinder connecting map here is the negative of the one in Lecture 5. Homotopy invariance makes $\theta_{h,*}$ invertible, so transport that sequence to $A_n(Z)$. The identities on the two restrictions of θ_h identify the adjacent homomorphism with $j_* - i_*$. Naturality follows for compatible maps of the chosen squares, as in Lecture 5. $\square$

Remark 3.2. Degree-wise homology groups need not form a pushout square in $\mathbf{Ab}$. The ordinary pushout of $A_n(X) \leftarrow A_n(W) \rightarrow A_n(Y)$ is canonically isomorphic to the cokernel of (f_*, g_*) , by changing the sign on the $A_n(X)$ summand. Exactness in Proposition 3.1 identifies this cokernel with $\ker(\delta_n^Z)$, a subgroup of $A_n(Z)$. It is all of $A_n(Z)$ if and only if $\delta_n^Z = 0$. For the suspension square of S^0 in Example 1.4, ordinary H_1 vanishes at the other three corners but $H_1(S^1) = \mathbb{Z}$. The preservation statement is the long exact sequence, or the derived pushout of chains in Proposition 2.5.

For a nonempty pointed space (X, x_0) , define reduced generalized homology by

$$\tilde{A}_n(X) = \ker(A_n(X) \xrightarrow{(\text{pr}_X)_*} A_n(*)),$$

where $\text{pr}_X : X \rightarrow *$ is the constant map. The basepoint inclusion $e : * \rightarrow X$ satisfies $\text{pr}_X e = \text{id}_*$ and splits this map. Consequently $A_n(X) = \tilde{A}_n(X) \oplus A_n(*)$. Pointed maps preserve the splitting. This definition applies to H_n as well, with $H_n(*) = 0$ for $n \neq 0$ and $H_0(*) = \mathbb{Z}$.

Proposition 3.3. Let W be a nonempty pointed topological space. For its unreduced suspension SW , pointed at the top vertex, the connecting map gives isomorphisms

$$\tilde{A}_n(SW) \xrightarrow{\cong} \tilde{A}_{n-1}(W) \quad (n \in \mathbb{Z}).$$

Proof. Apply Proposition 3.1 to $* \xleftarrow{\text{pr}_W} W \xrightarrow{\text{pr}_W} *$. In degree $n-1$, the map from $A_{n-1}(W)$ to the two point groups is $((\text{pr}_W)_*, (\text{pr}_W)_*)$, whose kernel is $\tilde{A}_{n-1}(W)$. Exactness makes this kernel the image of δ_n .

The map from the two point groups into $A_n(SW)$ is the second vertex map minus the first. These vertices are joined by the interval over the basepoint of W , so their maps on homology agree. The image of their difference on the direct sum is the coefficient summand $A_n(*)$ in $A_n(SW)$. Exactness therefore identifies $A_n(SW)/A_n(*)$ with $\tilde{A}_{n-1}(W)$. The basepoint splitting identifies this quotient with $\tilde{A}_n(SW)$, proving the assertion. $\square$

For singular homology, additivity gives $\tilde{H}_0(S^0) = \mathbb{Z}$ and $\tilde{H}_n(S^0) = 0$ for $n \neq 0$. Iterating Proposition 3.3 along $SS^m \cong S^{m+1}$ yields $\tilde{H}_n(S^m) = \mathbb{Z}$ when $n = m$ and zero otherwise. For generalized homology the same argument gives $\tilde{A}_n(S^m) \cong A_{n-m}(*)$; the coefficient groups need not be concentrated in degree zero.

3.2 Cell Attachments and Mapping Cones

The exact sequence retains both the gluing maps and the connecting map, so it determines which classes survive an attachment and which new classes occur.

Example 3.4. Cover S^1 by the complements U, V of two distinct points. Each is an open interval and the intersection has two interval components. The sequence of Theorem 2.2 contains

$$0 \rightarrow H_1(S^1) \xrightarrow{\delta_1} \mathbb{Z}^2 \xrightarrow{(f_*, g_*)} \mathbb{Z}^2 \rightarrow H_0(S^1) \rightarrow 0.$$

Each component of $U \cap V$ maps to the unique component of each cover member. Thus the middle map sends (u, v) to $(u + v, u + v)$. Its kernel is generated by $(1, -1)$, so $H_1(S^1) \cong \mathbb{Z}$. Its cokernel is $\mathbb{Z}$, in agreement with connectedness. Splitting an oriented circuit into its two arcs gives a boundary equal to the difference of the two overlap points; this is the formula for δ_1 in Theorem 2.2.

Proposition 3.5. Let $m \geq 2$ and let $f : S^{m-1} \rightarrow X$ attach an m -cell, producing $Z = X \underset{S^{m-1}}{+} D^m$. There is an exact sequence

$$0 \rightarrow H_m(X) \rightarrow H_m(Z) \rightarrow \mathbb{Z} \xrightarrow{f_*} H_{m-1}(X) \rightarrow H_{m-1}(Z) \rightarrow 0.$$

The homomorphism f_* sends a chosen generator of $H_{m-1}(S^{m-1})$ to the homology class of the attaching map. In positive degrees other than $m, m-1$, the inclusion induces $H_n(X) \cong H_n(Z)$; it also induces an isomorphism on H_0 .

Proof. Example 1.7 gives a homotopy pushout. Apply Theorem 2.3, using the homology of the sphere from Proposition 3.3 and the contractibility of D^m . In the relevant positive degrees, the disk groups vanish and $H_{m-1}(S^{m-1}) = \mathbb{Z}$, giving the displayed sequence. The remaining positive-degree terms vanish on the sphere and disk, giving the stated isomorphisms.

Since $m \geq 2$, the attaching sphere is connected. In degree zero the map $H_0(S^{m-1}) \rightarrow H_0(X) \oplus H_0(D^m)$ identifies the disk component with the component of X containing the image of f . Its cokernel is $H_0(X)$, proving the final assertion. $\square$

Example 3.6. Attach an m -cell to S^{m-1} along a map whose induced homomorphism on $H_{m-1}(S^{m-1})$ is multiplication by an integer d , with $m \geq 2$. Proposition 3.5 gives

$$H_m(S^{m-1} \underset{S^{m-1}}{+} D^m) = \ker(d : \mathbb{Z} \rightarrow \mathbb{Z}), \quad H_{m-1}(S^{m-1} \underset{S^{m-1}}{+} D^m) = \mathbb{Z}/d\mathbb{Z}.$$

For $d \neq 0$ these groups are 0 and $\mathbb{Z}/d\mathbb{Z}$; for $d = 0$ both are $\mathbb{Z}$. All other reduced homology groups vanish. For $m = 2$ and $d = 2$, this is the usual disk attachment describing $\mathbb{R}P^2$, with $H_1(\mathbb{R}P^2) = \mathbb{Z}/2$ and $H_2(\mathbb{R}P^2) = 0$.

For a pointed map $f : W \rightarrow X$ between nonempty pointed spaces, Example 1.8 and Proposition 3.1 also give the reduced cone sequence

$$\cdots \rightarrow \tilde{A}_n(W) \xrightarrow{f_*} \tilde{A}_n(X) \xrightarrow{-i_*} \tilde{A}_n(\text{Cone}(f)) \rightarrow \tilde{A}_{n-1}(W) \rightarrow \cdots.$$

Here $i : X \rightarrow \text{Cone}(f)$ is the canonical inclusion. To see the reduction, split off the point groups in the unreduced Mayer-Vietoris sequence. On those summands the maps are the split exact sequence $A_n(*) \xrightarrow{u \mapsto (u,u)} A_n(*)^2 \xrightarrow{(u,v) \mapsto v-u} A_n(*)$. The cone on the basepoint gives the homotopy identifying the two point inclusions, and its connecting map is zero. The remaining summands are the displayed reduced sequence. If f is a Hurewicz cofibration, it computes $\tilde{A}_n(X/W)$ by Proposition 1.6.

References. A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002: Proposition 2.21, pp. 119–123, and §2.2, pp. 149–150. J. P. May, *A Concise Course in Algebraic Topology*, University of Chicago Press, 1999: Chapter 6, §1, and Chapter 10, §§1,7, especially p. 80.